

Problem A. Commencement

Time Limit 2000 ms

Problem Statement

A string S consisting of lowercase English letters is a **good string** if and only if it satisfies the following property for all integers i not less than 1:

- There are exactly zero or exactly two different letters that appear exactly i times in S .

Given a string S , determine if it is a good string.

Constraints

- S is a string of lowercase English letters with a length between 1 and 100, inclusive.

Input

The input is given from Standard Input in the following format:

S

Output

Print **Yes** if S is a good string, and **No** otherwise.

Sample 1

Input	Output
commencement	Yes

For the string **commencement**, the number of different letters that appear exactly i times is as follows:

- $i = 1$: two letters (**o** and **t**)
- $i = 2$: two letters (**c** and **n**)
- $i = 3$: two letters (**e** and **m**)

- $i \geq 4$: zero letters

Therefore, **commencement** satisfies the condition of a good string.

Sample 2

Input	Output
banana	No

For the string **banana**, there is only one letter that appears exactly one time, which is **b**, so it does not satisfy the condition of a good string.

Sample 3

Input	Output
ab	Yes

Problem B. Counting Rooms

Time Limit 1000 ms
Mem Limit 524288 kB

You are given a map of a building, and your task is to count the number of its rooms. The size of the map is $n \times m$ squares, and each square is either floor or wall. You can walk left, right, up, and down through the floor squares.

Input

The first input line has two integers n and m : the height and width of the map.

Then there are n lines of m characters describing the map. Each character is either  (floor) or  (wall).

Output

Print one integer: the number of rooms.

Constraints

- $1 \leq n, m \leq 1000$

Example

Input	Output
5 8 ##### #..#...# ####.#.# #..#...# #####	3

Problem C. Grid Repainting

Time Limit 2000 ms

Problem statement

We have an $H \times W$ grid whose squares are painted black or white. The square at the i -th row from the top and the j -th column from the left is denoted as (i, j) .

Snuke would like to play the following game on this grid. At the beginning of the game, there is a character called Kenus at square $(1, 1)$. The player repeatedly moves Kenus up, down, left or right by one square. The game is completed when Kenus reaches square (H, W) passing only white squares.

Before Snuke starts the game, he can change the color of some of the white squares to black. However, he cannot change the color of square $(1, 1)$ and (H, W) . Also, changes of color must all be carried out before the beginning of the game.

When the game is completed, Snuke's score will be the number of times he changed the color of a square before the beginning of the game. Find the maximum possible score that Snuke can achieve, or print -1 if the game cannot be completed, that is, Kenus can never reach square (H, W) regardless of how Snuke changes the color of the squares.

The color of the squares are given to you as characters $s_{i,j}$. If square (i, j) is initially painted by white, $s_{i,j}$ is `.`; if square (i, j) is initially painted by black, $s_{i,j}$ is `#`.

Constraints

- H is an integer between 2 and 50 (inclusive).
- W is an integer between 2 and 50 (inclusive).
- $s_{i,j}$ is `.` or `#` ($1 \leq i \leq H, 1 \leq j \leq W$).
- $s_{1,1}$ and $s_{H,W}$ are `.`.

Input

Input is given from Standard Input in the following format:

```
H W
s1,1s1,2s1,3...s1,W
s2,1s2,2s2,3...s2,W
```

```

:      :
sH,1sH,2sH,3...sH,W

```

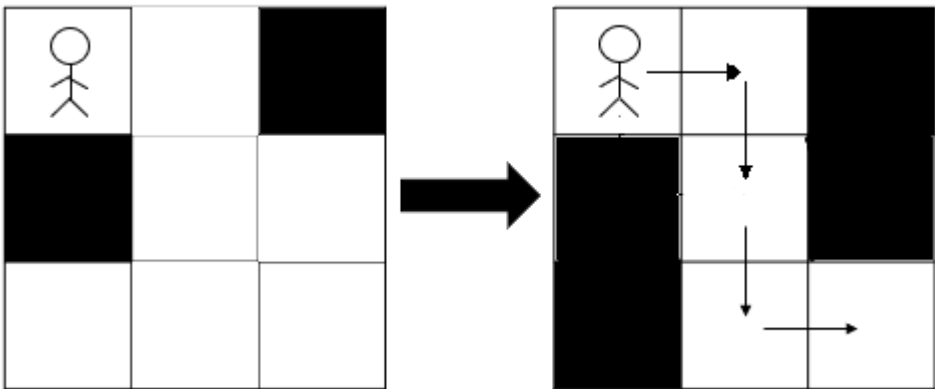
Output

Print the maximum possible score that Snuke can achieve, or print -1 if the game cannot be completed.

Sample 1

Input	Output
3 3 ..# #.. ...	2

The score 2 can be achieved by changing the color of squares as follows:



Sample 2

Input	Output
10 37#...####...####...###...###...###.. ..#.#...#...#.#...#...#.#...#...#.. ..#.#...#...#.#...#...#.#...#...#.. .#...#.#...##.#...#...#.#.###.#.###.. .#####.#####.#...#...#...##...##... .#...#.#...#.#...#...#.#...#.#...#.. .#...#.#...#.#...#...#.#...#.#...#.. .#...#.#...#.#...#...#.#...#.#...#.. .#...#.#...#.#...#...#.#...#.#...#..	209

Problem D. Slimes

Time Limit 2000 ms

Problem Statement

There are A slimes.

Each time Snuke shouts, the slimes multiply by K times.

In order to have B or more slimes, at least how many times does Snuke need to shout?

Constraints

- $1 \leq A \leq B \leq 10^9$
- $2 \leq K \leq 10^9$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

A B K

Output

Print the answer.

Sample 1

Input	Output
1 4 2	2

We start with one slime. After Snuke's first shout, we have two slimes; after his second shout, we have four slimes. Thus, he needs to shout at least twice to have four or more slimes.

Sample 2

Input	Output
7 7 10	0

We have seven slimes already at the start.

Sample 3

Input	Output
31 415926 5	6

Problem E. Maze Master

Time Limit 2000 ms

Problem Statement

Takahashi has a maze, which is a grid of $H \times W$ squares with H horizontal rows and W vertical columns.

The square at the i -th row from the top and the j -th column is a "wall" square if S_{ij} is `#`, and a "road" square if S_{ij} is `.`.

From a road square, you can move to a horizontally or vertically adjacent road square.

You cannot move out of the maze, move to a wall square, or move diagonally.

Takahashi will choose a starting square and a goal square, which can be any road squares, and give the maze to Aoki.

Aoki will then travel from the starting square to the goal square, in the minimum number of moves required.

In this situation, find the maximum possible number of moves Aoki has to make.

Constraints

- $1 \leq H, W \leq 20$
- S_{ij} is `.` or `#`.
- S contains at least two occurrences of `.`.
- Any road square can be reached from any road square in zero or more moves.

Input

Input is given from Standard Input in the following format:

```
H W
S11...S1W
:
SH1...SHW
```


Output

Print the maximum possible number of moves Aoki has to make.

Sample 1

Input	Output
3 3	4

If Takahashi chooses the top-left square as the starting square and the bottom-right square as the goal square, Aoki has to make four moves.

Sample 2

Input	Output
3 5 ...#. .#.#. .#...	10

If Takahashi chooses the bottom-left square as the starting square and the top-right square as the goal square, Aoki has to make ten moves.

Problem F. World Cup

Time Limit 2000 ms

Problem Statement

A sport event is held in June of every year whose remainder when divided by 4 is 2.

Suppose that it is now January of the year Y . In what year will this sport event be held next time?

Constraints

- $2000 \leq Y \leq 3000$
- Y is an integer.

Input

Input is given from Standard Input in the following format:

Y

Output

Print the answer.

Sample 1

Input	Output
2022	2022

The remainder when 2022 is divided by 4 is 2, so if it is now January of 2022, the next games will be held in June of the same year.

Sample 2

Input	Output
2023	2026

Sample 3

Input	Output
3000	3002

Problem G. Darker and Darker

Time Limit 1000 ms

Problem Statement

You are given a grid of squares with H horizontal rows and W vertical columns, where each square is painted white or black. HW characters from A_{11} to A_{HW} represent the colors of the squares. A_{ij} is `#` if the square at the i -th row from the top and the j -th column from the left is black, and A_{ij} is `.` if that square is white.

We will repeatedly perform the following operation until all the squares are black:

- Every white square that shares a side with a black square, becomes black.

Find the number of operations that will be performed. The initial grid has at least one black square.

Constraints

- $1 \leq H, W \leq 1000$
- A_{ij} is `#` or `.`
- The given grid has at least one black square.

Input

Input is given from Standard Input in the following format:

```
H W
A11A12...A1W
:
AH1AH2...AHW
```

Output

Print the number of operations that will be performed.

Sample 1

Input	Output
3 3#. ...	2

After one operation, all but the corners of the grid will be black. After one more operation, all the squares will be black.

Sample 2

Input	Output
6 6 ..#...# #...#..#.....#..	3

Problem H. Labyrinth

Time Limit 1000 ms

Mem Limit 524288 kB

You are given a map of a labyrinth, and your task is to find a path from start to end. You can walk left, right, up and down.

Input

The first input line has two integers n and m : the height and width of the map.

Then there are n lines of m characters describing the labyrinth. Each character is `.` (floor), `#` (wall), `A` (start), or `B` (end). There is exactly one `A` and one `B` in the input.

Output

First print "YES", if there is a path, and "NO" otherwise.

If there is a path, print the length of the shortest such path and its description as a string consisting of characters `L` (left), `R` (right), `U` (up), and `D` (down). You can print any valid solution.

Constraints

- $1 \leq n, m \leq 1000$

Example

Input	Output
<pre> 5 8 ##### #.A#...# #.#.#B# #.....# ##### </pre>	<pre> YES 9 LDDRRRRRU </pre>

Problem I. Knight Moves Grid

Time Limit 1000 ms
Mem Limit 524288 kB

There is a knight on an $n \times n$ chessboard. For each square, print the minimum number of moves the knight needs to do to reach the top-left corner.

Input

The only line has an integer n .

Output

Print the number of moves for each square.

Constraints

- $4 \leq n \leq 1000$

Example

Input	Output
8	0 3 2 3 2 3 4 5 3 4 1 2 3 4 3 4 2 1 4 3 2 3 4 5 3 2 3 2 3 4 3 4 2 3 2 3 4 3 4 5 3 4 3 4 3 4 5 4 4 3 4 3 4 5 4 5 5 4 5 4 5 4 5 6

Problem J. Monsters

Time Limit 1000 ms

Mem Limit 524288 kB

You and some monsters are in a labyrinth. When taking a step to some direction in the labyrinth, each monster may simultaneously take one as well. Your goal is to reach one of the boundary squares without ever sharing a square with a monster.

Your task is to find out if your goal is possible, and if it is, print a path that you can follow. Your plan has to work in any situation; even if the monsters know your path beforehand.

Input

The first input line has two integers n and m : the height and width of the map.

After this there are n lines of m characters describing the map. Each character is `.` (floor), `#` (wall), `A` (start), or `M` (monster). There is exactly one `A` in the input.

Output

First print "YES" if your goal is possible, and "NO" otherwise.

If your goal is possible, also print an example of a valid path (the length of the path and its description using characters `D`, `U`, `L`, and `R`). You can print any path, as long as its length is at most $n \cdot m$ steps.

Constraints

- $1 \leq n, m \leq 1000$

Example

Input	Output
<pre> 5 8 ##### #M..A..# #.#.M#.# #M#...# #.#.##### </pre>	<pre> YES 5 RRDDR </pre>